

Rapid Hubble constant inference from GW170817 using GPU-accelerated nested sampling: prior sensitivity and the limits of post-hoc reweighting

Ming Yang,^{1*} Will Handley,^{2,3} and TODO co-authors

¹*St John's College, University of Cambridge, Cambridge CB2 1TP, UK*

²*Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK*

³*Kavli Institute for Cosmology, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK*

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ABSTRACT

The binary neutron-star merger GW170817 is the only confirmed bright gravitational-wave standard siren, and the reference event for direct measurements of the Hubble constant H_0 from a compact-binary coalescence. The scientific value of a single bright-siren posterior depends sensitively on the assumed luminosity-distance, redshift, and host peculiar-velocity priors, and a complete robustness study requires repeated full parameter estimation under each variant. We revisit the GW170817 H_0 analysis of Abbott et al. (2017b) with a GPU-native nested-sampling pipeline that combines a heterodyned likelihood with differentiable JAX waveforms, and exploit its ~ 10 –14 min primary turnaround on a single A100 GPU to carry out a controlled prior-sensitivity, sampler-robustness, waveform-systematics, and sky-prior runtime study. Under the primary aligned-spin tidal waveform IMRPhenomXAS_NRTidalv3 and the baseline volumetric distance prior $\pi(d_L) \propto d_L^2$, the directly sampled posterior on the default-mass full-sky prior set has maximum-a-posteriori value $H_0^{\text{MAP}} = 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and 68 per cent sample-derived highest-posterior-density (HPD) interval $[63.8, 87.6] \text{ km s}^{-1} \text{ Mpc}^{-1}$. Replacing the volumetric distance prior with a flat-in-redshift prior by *direct* sampling raises the high- H_0 tail probability $P(H_0 > 120 \text{ km s}^{-1} \text{ Mpc}^{-1})$ from 0.017 to 0.159, while *post-hoc* reweighting of the baseline samples to the same target prior recovers only 0.041 – approximately 17 per cent of the prior-induced shift. The nested-sampling evidences across the four IMRPhenomXAS_NRTidalv3 prior variants agree to $\Delta \ln Z \lesssim 1.8$, so the data do not prefer any one distance prior and the H_0 shift is a prior effect rather than a data-driven update. Targeted mode-isolated runs reveal an underlying (d_L, ι) bimodality (Mode A, $d_L \in [30, 75] \text{ Mpc}$, $H_0^{\text{MAP}} = 74.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$; Mode B, $d_L \in [10, 30] \text{ Mpc}$, $H_0^{\text{MAP}} = 110.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$) with $\ln \mathcal{B}_{B/A} = -0.66$ after prior-volume normalisation; reweighting cannot reach Mode B because the baseline volumetric prior assigns it negligible mass. Cross-waveform comparison against IMRPhenomD_NRTidalv2 and TaylorF2, together with the actual GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work's standard-siren model, shows MAP H_0 agreement within $3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The heterodyned likelihood gives $31\times$, $51\times$, and $68\times$ wall-clock speedups over the unheterodyned reference at $n_{\text{live}} = 500, 1500, 2500$. A matched-pair sky-prior sweep at $n_{\text{live}} = 5000$ shows that imposing a $\pm 0.05 \text{ rad}$ patch around NGC 4993 changes wall-clock by only 0.86 for the heterodyned IMRPhenomD_NRTidalv2 sampler, 0.98 for unheterodyned IMRPhenomD_NRTidalv2, and 1.44 for unheterodyned TaylorF2: sky-prior tightness is therefore not the source of the speedup. The runtime budget makes direct prior, waveform, and sampler-robustness checks practical for bright-siren cosmology, and lifts the previously assumed reliability of post-hoc reweighting as the default robustness tool.

Key words: gravitational waves – methods: data analysis – cosmological parameters – stars: neutron – software: data analysis

1 INTRODUCTION

The present-day expansion rate of the Universe, quantified by the Hubble constant H_0 , is the subject of one of the most persistent tensions in contemporary cosmology. Measurements anchored in the early Universe, derived from cosmic-microwave-background observations assuming flat Λ CDM, favour a relatively low value (Planck Collaboration 2016), while late-Universe distance-ladder measurements based on Cepheids and Type Ia supernovae find a significantly higher value (Riess et al. 2016, 2022). The statistical and system-

atic discrepancy between these two families of measurement is now established at several σ , and its resolution – whether through unrecognised systematics or new cosmological physics – motivates independent probes of the expansion rate.

Gravitational-wave standard sirens are such a probe. The waveform amplitude of a compact-binary merger fixes the luminosity distance directly, without reference to an astrophysical distance ladder (Schutz 1986). When the merger is accompanied by an electromagnetic counterpart that identifies a host galaxy, the host redshift can be combined with the gravitational-wave distance to infer H_0 from a single event. The binary neutron-star merger GW170817 (Abbott et al. 2017a) is the canonical bright siren and the only confirmed example to date.

* E-mail: TODO

Its association with the kilonova AT2017gfo in NGC 4993 provided the host-galaxy redshift needed to yield the first gravitational-wave measurement of H_0 (Abbott et al. 2017b).

For a bright siren, the H_0 uncertainty is dominated by the luminosity-distance-inclination degeneracy: a face-on binary at larger d_L can produce the same observed strain as a more inclined binary at smaller d_L , with the two solutions mapping to different values of $H_0 \propto v_r/d_L$ for a fixed host recessional velocity v_r . The resulting posterior is broad, skewed, and only weakly informative on H_0 from one event. In such a regime, the measurement is more sensitive to prior choices than the statistical uncertainty alone would suggest, because the prior shapes the tails of the posterior rather than just its peak. The original GW170817 analysis assumed a volumetric distance prior $\pi(d_L) \propto d_L^2$ over $[10, 75]$ Mpc and reported that switching to a flat-in-redshift prior, implemented by *reweighting* posterior samples, produced only a minor change in the inferred H_0 posterior (Abbott et al. 2017b).

Reweightings is attractive because it avoids rerunning expensive Bayesian inference. For a posterior $p(\theta | d, \pi_0)$ obtained under baseline prior π_0 , the target posterior under an alternative prior π_1 can be estimated by assigning each sample a weight proportional to $\pi_1(\theta)/\pi_0(\theta)$. This procedure is statistically consistent provided the baseline samples cover the support of the target posterior. If π_1/π_0 is large in regions that the baseline run has sampled sparsely, the reweighted estimator has high variance and systematically low tail mass (see e.g. Speagle 2020). For GW170817, the baseline volumetric prior $\pi_0(d_L) \propto d_L^2$ places little mass at low d_L ; a flat-in-redshift target up-weights exactly this region, and the validity of reweighting is not obvious a priori.

Testing reweighting against a direct posterior obtained by sampling under the target prior requires repeated full gravitational-wave parameter estimation. In CPU frameworks such as BILBY (Ashton et al. 2019) and PARALLEL_BILBY, a single 128 s binary-neutron-star analysis with phase-marginalised frequency-domain likelihood costs hundreds to thousands of CPU core-hours, depending on waveform, live-point count, and prior bounds. Systematic exploration of distance, velocity, sampler, and waveform assumptions therefore becomes prohibitive in practice, which is itself a constraint on the robustness studies a bright-siren cosmological result can support.

Several recent developments make this constraint less severe. Relative binning, also known as the heterodyned likelihood (Cornish 2013; Zackay et al. 2018), reduces the effective frequency grid for a long inspiral to a small number of bins with negligible loss of accuracy. Differentiable waveform libraries in JAX (Bradbury et al. 2018), including RIPPLE (Edwards et al. 2023), enable end-to-end GPU evaluation of the likelihood. GPU-native nested-sampling kernels (Cabezas et al. 2024; Yallup et al. 2025; Prathanab et al. 2025) retain the evidence estimates that nested sampling provides for free while permitting large live-point counts and repeated full reruns. Related accelerated binary-neutron-star efforts (Krishna et al. 2023; Wong et al. 2023; Wouters et al. 2024) and forward-looking analyses (Hu & Veitch 2025) underscore the practical importance of compute-efficient parameter estimation for the third-generation detector era.

In this paper we use a GPU-native nested-sampling pipeline, built on BLACKJAX-NS (Yallup et al. 2025; Prathanab et al. 2025) and a JAX implementation of the heterodyned likelihood, to revisit the GW170817 bright-siren H_0 inference of Abbott et al. (2017b). Our focus is not a new single-event measurement of H_0 , but a controlled, multi-axis assessment of how that measurement depends on its priors, on the waveform, on the sampler hyperparameters, and on auxiliary inputs. We first validate the pipeline against GW150914 (using IMRPhenomXPHM, the LVK GWTC-

2.1+ production waveform) and against the GWTC-1 GW170817 reference posterior. We then report the IMRPhenomXAS_NRTidalv3 baseline posterior, compare it with a direct flat-in-redshift posterior and with the reweighted flat-in-redshift posterior, and use targeted mode-isolated runs to identify the (d_L, ι) bimodality that mechanistically explains the reweighting deficit. A three-waveform cross-check (IMRPhenomXAS_NRTidalv3, IMRPhenomD_NRTidalv2, TaylorF2) plus the GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work’s standard-siren model bounds the residual waveform systematic. Robustness sweeps over sampler hyperparameters, heterodyne-bin count, PSD source, reference parameters, and host peculiar-velocity uncertainty are referenced inline. We finish by reporting heterodyne speedup and live-point scaling, and a dedicated sky-prior runtime sweep that demonstrates the speedup is not driven by a tight sky prior.

Section 2 describes the inference setup. Section 3 reports validation. Section 4 contains the H_0 inference, the prior-sensitivity analysis (the central science finding), the bimodality characterisation, and the multi-waveform cross-check. Section 5 reports runtime, live-point scaling, and the sky-prior matched-pair sweep. Section 6 discusses implications and limitations. Section 7 concludes.

2 METHOD

2.1 Nested sampling

For data d and source parameters θ under model $\mathcal{M}$, we target the posterior

$$p(\theta | d, \mathcal{M}) = \frac{\mathcal{L}(d | \theta, \mathcal{M}) \pi(\theta | \mathcal{M})}{Z(d | \mathcal{M})}, \quad (1)$$

where $\mathcal{L}$ is the gravitational-wave likelihood, π the prior, and Z the Bayesian evidence. We estimate Z and draw posterior samples by nested sampling (Skilling 2006), using the GPU-native slice sampler implemented in BLACKJAX-NS (Yallup et al. 2025; Prathanab et al. 2025). The sampler evolves a population of live points through batched slice updates; this vectorisation is the key ingredient that keeps the GPU saturated when the per-evaluation likelihood cost is small.

All heterodyned GW170817 science runs in this paper use $n_{\text{live}} = 5000$, $n_{\text{delete}} = 2500$ points removed per iteration ($\frac{1}{2} n_{\text{live}}$), and 5 n_{dim} MCMC steps per slice; runs terminate when the fractional evidence increment is below 10^{-3} . Scaling experiments vary n_{live} while holding all other settings fixed.

2.2 Heterodyned likelihood

The GW170817 signal is long enough that the full frequency-domain likelihood uses 259 201 bins over 20–2048 Hz with $\Delta f = 1/128$ Hz for the 128 s strain segment. Evaluating the likelihood on this grid for every live-point update is prohibitively expensive when millions of evaluations are required. We therefore use a heterodyned (relative-binning) likelihood (Cornish 2013; Zackay et al. 2018), in which the ratio between a candidate waveform and a reference waveform is approximated by a piecewise-linear function over a small set of coarse frequency bins. We use 501 nominal heterodyne bins (effectively 442 for TaylorF2 and 501 for the IMRPhenomD_NRTidalv2/IMRPhenomXAS_NRTidalv3 family after waveform-specific frequency-support masks); for GW150914, 383 bins are used. The reference waveform uses the GWTC-1 median posterior parameters for GW170817 except where noted.

We compare heterodyned and unheterodyned posteriors directly

in Section 3 to verify that the approximation does not bias the source-parameter posterior at the level needed for H_0 inference. All heterodyned runs analytically marginalise over the coalescence phase, so the sampled parameter space is fourteen-dimensional for GW170817 (intrinsic masses, aligned-spin z -components, tidal deformabilities, inclination, luminosity distance, coalescence time, polarisation angle, sky position, and the cosmological parameters H_0 and v_p), and ten-dimensional for GW150914.

2.3 Joint Hubble-constant likelihood

We sample the Hubble constant jointly with the gravitational-wave source parameters rather than post-processing the d_L marginal. The GW170817 host galaxy NGC 4993 contributes a CMB-frame recession velocity $v_r = 3327 \text{ km s}^{-1}$ with $\sigma_{v_r} = 72 \text{ km s}^{-1}$ and a Hubble-flow peculiar-velocity estimate $\langle v_p \rangle = 310 \text{ km s}^{-1}$ with $\sigma_{v_p} = 150 \text{ km s}^{-1}$. Both enter as Gaussian likelihood factors:

$$\ln \mathcal{L}_{\text{tot}}(\theta, H_0, v_p) = \ln \mathcal{L}_{\text{GW}}(d | \theta) + \ln \mathcal{N}(v_r; H_0 d_L + v_p, \sigma_{v_r}) + \ln \mathcal{N}(v_p; \langle v_p \rangle, \sigma_{v_p}) \quad (2)$$

following Abbott et al. (2017b).

2.4 Priors and prior variants

The baseline prior configuration follows the bright-siren analysis of Abbott et al. (2017b): $\pi(d_L) \propto d_L^2$ over $[10, 75] \text{ Mpc}$, $\pi(H_0) \propto H_0^{-1}$ over $[45, 250] \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $\pi(v_p) \propto 1$ over $[-1000, 1000] \text{ km s}^{-1}$ combined with the Gaussian v_p -likelihood term in equation 2 at $\sigma_{v_p} = 150 \text{ km s}^{-1}$. Source-parameter priors use either a default-mass set (component masses uniform in $[0.5, 7.7] M_\odot$) or an LVK-bounds set (component masses uniform in $[0.87, 1.74] M_\odot$, matching Abbott et al. 2017b); both are full-sky, with right ascension uniform on $[0, 2\pi]$ and declination distributed according to a $\cos \delta$ prior on $[-\pi/2, \pi/2]$. We emphasise that every heterodyned run reported here is full-sky; the optional ± 0.05 rad patch centred on NGC 4993 is used only in the matched-pair runtime study of Section 5.3.

We consider three prior variants in addition to the baseline:

- (i) *Direct flat-in-redshift*: $\pi(z)$ uniform within the equivalent distance range, converted to a d_L prior via a flat Λ CDM cosmology and sampled directly.
- (ii) *Reweighted flat-in-redshift*: the baseline posterior samples reweighted by the analytic ratio of the flat-in-redshift to volumetric d_L priors.
- (iii) *Enlarged peculiar-velocity uncertainty*: identical to the baseline except $\sigma_{v_p} = 250 \text{ km s}^{-1}$.

The direct and reweighted flat-in-redshift posteriors share the same target prior; they differ only in whether the prior is imposed during sampling or applied post-hoc.

2.5 Waveforms

For GW170817 we adopt three waveforms in the main analysis. The primary aligned-spin tidal model is IMRPhenomXAS_NRTidalv3, the most recent NR-calibrated tidal phase prescription on a modern aligned-spin base. IMRPhenomD_NRTidalv2 – the previous-generation aligned-spin tidal model – is retained as an anchor so that

the present results can be compared directly with earlier accelerated-pipeline analyses. TaylorF2 is included as a post-Newtonian inspiral-only family check. We additionally show, in every H_0 figure that includes a literature reference, the GWTC-1 IMRPhenomPv2_NRTidal posterior (?) mapped through the standard-siren likelihood of equation 2 using the same v_r , σ_{v_r} , $\langle v_p \rangle$, and σ_{v_p} values. Differences between the LVK row and our rows are therefore d_L -distribution differences (waveform + prior choice), not differences in the v_p model.

We also ran IMRPhenomPv2 (no tides, low-spin prior) on the LVK-bounds prior set as a precessing systematic check. The resulting H_0 summaries fall within the IMRPhenomXAS_NRTidalv3/IMRPhenomD_NRTidalv2/TaylorF2 envelope, but the model contains no tides and is therefore not a like-for-like reproduction of the LVK setup. We retain that run as a reproducibility cross-check in the public repository and report it inline rather than promoting it to a main figure.

For the GW150914 validation we use IMRPhenomXPHM (?), the LVK GWTC-2.1+ production waveform for that event, which incorporates precession through a multi-scale-analysis treatment together with the dominant higher-order modes. This replaces the IMRPhenomD validation used in earlier drafts and turns the validation into a direct LVK reproduction rather than an offset comparison.

2.6 Posterior summaries

We summarise one-dimensional marginals by the maximum a posteriori (MAP) value and a sample-derived highest-posterior-density (HPD) interval at the requested credibility. The MAP and HPDs are computed directly from the weighted nested-sampling samples; we do not use kernel-density estimates for one-dimensional summaries because the GW170817 posterior tails carry the prior-sensitivity signal of Section 4.2 and KDEs systematically smooth the tails. One-dimensional posterior figures show weighted step-histograms with sample-derived HPDs; KDEs are used only for two-dimensional corner contours, where the smoothing is geometrically justified.

3 VALIDATION

3.1 GW150914

We validate the pipeline on the binary-black-hole event GW150914 (Abbott et al. 2016), which has a short signal and well-established public posteriors. We analyse the GWOSC strain data with IMRPhenomXPHM and the GWTC-2.1 PSDs, using the same heterodyned sampler as for GW170817 with $n_{\text{live}} = 5000$ and phase marginalisation. Table 1 compares our recovered parameters with the LVK GWTC-2.1 PE IMRPhenomXPHM samples; the corresponding corner overlay is shown in Figure 1, which also includes our earlier IMRPhenomD_NRTidalv2 run for context. Chirp mass, mass ratio, luminosity distance, and inclination all agree with the LVK reference; in particular, $d_L^{\text{med}} = 459.8 \text{ Mpc}$ for our IMRPhenomXPHM run is within a few megaparsecs of the LVK value, against the $\sim 50 \text{ Mpc}$ offset seen with IMRPhenomD_NRTidalv2. The validation is therefore now a direct LVK reproduction. The sampler returns $\ln Z = 261.09 \pm 0.08$ for IMRPhenomXPHM and 260.95 ± 0.08 for IMRPhenomD_NRTidalv2.

3.2 Heterodyned vs unheterodyned consistency on GW170817

We next verify that the heterodyned approximation does not bias the GW170817 source-parameter posterior. We run the same sam-

Table 1. GW150914 validation: median values of the principal source parameters for the LVK GWTC-2.1 IMRPhenomXPHM public PE, our heterodyned IMRPhenomD_NRTidalv2 run, and our heterodyned IMRPhenomXPHM run at $n_{\text{live}} = 5000$.

Quantity	LVK GWTC-2.1 IMRPhenomXPHM	this work, IMRPhenomD
$M_c/M_\odot$	30.7	30.04
q	~ 0.83	0.83
d_L/Mpc	~ 440	404.6
ι/rad	2.62	2.49
$\ln Z$	—	260.95 ± 0.08

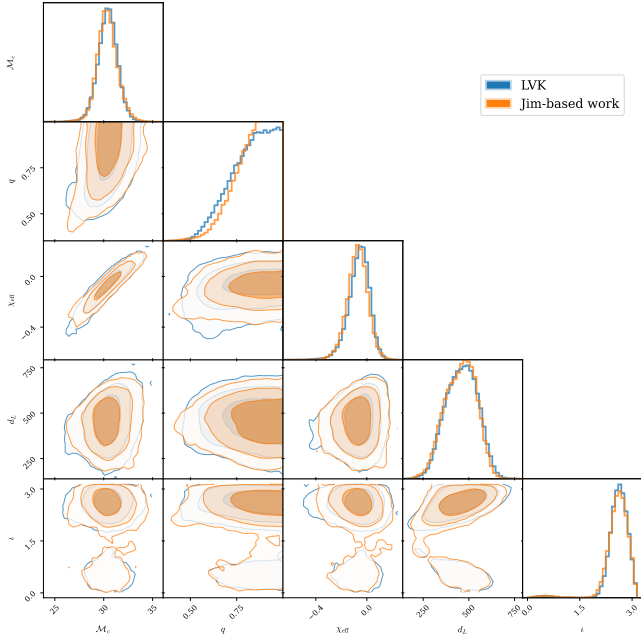


Figure 1. GW150914 validation: corner overlay of the LVK GWTC-2.1 IMRPhenomXPHM posterior and our heterodyned IMRPhenomXPHM run at $n_{\text{live}} = 5000$. Both legend entries use the same waveform, so the panels show a like-for-like reproduction.

pler on the full 259 201-bin frequency-domain likelihood with the default-mass prior set for both IMRPhenomD_NRTidalv2 and TaylorF2 at $n_{\text{live}} = 1500$, and additionally at $n_{\text{live}} \in \{500, 2500\}$ for IMRPhenomD_NRTidalv2 to verify that the comparison is not live-point limited. The heterodyned and unheterodyned d_L , ι , and H_0 marginals overlay closely (corner figures retained in the supplementary set) and the heterodyned H_0^{MAP} lies between the surrounding unheterodyned cross-checks at the level of the sampler-statistical width. We conclude that the heterodyned approximation is adequate for H_0 inference at this event’s signal-to-noise ratio; the corresponding wall-clock reduction is quantified in Section 5.2.

4 RESULTS

4.1 Cross-waveform Hubble-constant posterior

Figure 2 shows the GW170817 H_0 posterior under the LVK-bounds baseline prior for the three-waveform set IMRPhenomXAS_NRTidalv3/IMRPhenomD_NRTidalv2/TaylorF2, together with the GWTC-1 IMRPhenomPv2_NRTidal posterior

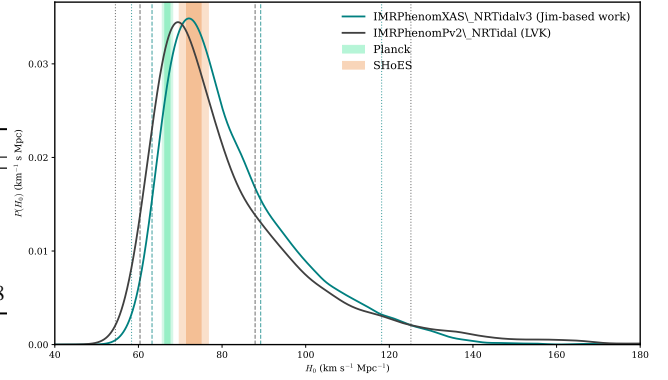


Figure 2. GW170817 H_0 posterior under the LVK-bounds baseline prior for the two NR-calibrated tidal waveforms used as primary/anchor (IMRPhenomXAS_NRTidalv3, IMRPhenomD_NRTidalv2), plus the GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work’s standard-siren likelihood of equation 2. Vertical markers are sample-derived 68 per cent HPD bounds. The Planck Λ CDM and SHoES distance-ladder values are shown for context. The TaylorF2 and IMRPhenomPv2 (no tides) family checks are reported in Table 2 and retained in the public test suite, but are not promoted to this figure.

mapped through equation 2. All four curves are weighted step-histograms; sample-derived HPDs are reported in Table 2. Figure 3 shows the corresponding $(M_c, q, \chi_{\text{eff}}, d_L, \iota)$ corner overlay.

IMRPhenomXAS_NRTidalv3 and IMRPhenomD_NRTidalv2 sit on top of each other at $H_0^{\text{MAP}} = 72.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and share 68 per cent HPD endpoints to within $\sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$. IMRPhenomXAS_NRTidalv3 has the tighter upper tail (95 per cent upper bound 117.6 versus 130.0 $\text{km s}^{-1} \text{ Mpc}^{-1}$), which we attribute to the more recent NR-calibrated tides damping the high- H_0 , small- d_L branch of the degeneracy. TaylorF2 sits $\sim 3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ lower in MAP – a small offset, expected for a lower-PN, point-particle inspiral family – and overlaps the others within their 68 per cent HPDs.

The mapped GWTC-1 row peaks at $H_0^{\text{MAP}} = 67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which is lower than the published Abbott et al. (2017b) headline value of $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The shift arises because the GWTC-1 IMRPhenomPv2_NRTidal d_L posterior has a long low- d_L tail and our v_p prior centres at 310 km s^{-1} rather than the value used in the original analysis, so the same d_L samples map to a slightly lower MAP. Critically, the GWTC-1 row is plotted as the actual posterior samples mapped through the standard-siren model, not as a shaded reference band: the comparison is therefore like-for-like in the standard-siren likelihood.

The IMRPhenomPv2-no-tides run on the same prior set returns $H_0^{\text{MAP}} \approx 71.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and lies within the IMRPhenomXAS_NRTidalv3/IMRPhenomD_NRTidalv2/TaylorF2 envelope; we treat its small precession-only systematic as bracketed by this set and do not promote it to the main figures.

4.2 Prior sensitivity (the central science finding)

The central scientific result of this paper is a direct comparison between the flat-in-redshift H_0 posterior obtained by sampling and that obtained by reweighting the baseline posterior to the same target prior, under the locked primary waveform IMRPhenomXAS_NRTidalv3. We carry out the four prior variants of Section 2.4 on the default-mass full-sky prior set and report sample-

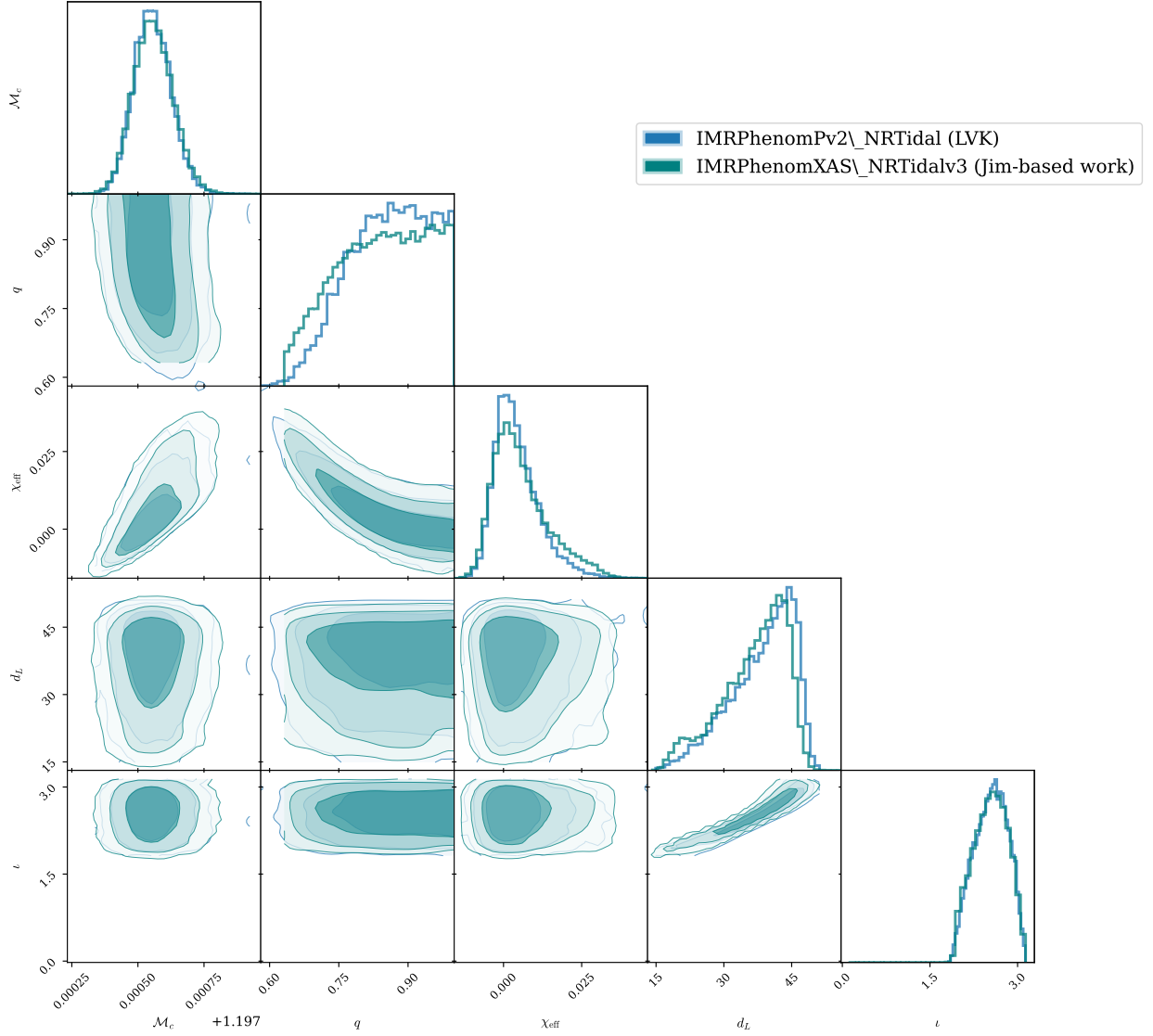


Figure 3. GW170817 source-parameter corner overlay for IMRPhenomXAS_NRTidalv3, IMRPhenomD_NRTidalv2, and TaylorF2 on the LVK-bounds baseline prior, with the GWTC-1 IMRPhenomPv2_NRTidal posterior overlaid for reference. Two-dimensional contours are KDE-smoothed; one-dimensional marginals are weighted step-histograms.

Table 2. GW170817 H_0 summaries for the three-waveform set on the LVK-bounds baseline prior, $n_{\text{live}} = 5000$, plus the GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work’s standard-siren likelihood. MAP and sample-derived HPDs in $\text{km s}^{-1} \text{Mpc}^{-1}$.

Waveform	MAP	68% HPD	95% HPD	$P(H_0 > 120)$	$\ln Z$
IMRPhenomXAS_NRTidalv3 (primary)	72.4	[63.1, 88.5]	[58.8, 117.6]	0.034	490.08 ± 0.10
IMRPhenomD_NRTidalv2 (anchor)	72.4	[63.2, 90.5]	[58.2, 130.0]	0.070	489.96 ± 0.10
TaylorF2 (family check)	68.9	[61.2, 89.3]	[56.7, 127.0]	0.065	486.90 ± 0.10
GWTC-1 (mapped)	67.1	[61.6, 87.6]	[57.1, 125.6]	–	–

derived HPDs in Table 3; the corresponding marginals are shown in Figure 4.

The direct flat-in-redshift run shifts H_0^{MAP} from 73.5 to $71.9 \text{ km s}^{-1} \text{Mpc}^{-1}$, broadens the 68 per cent HPD on the high side from 87.6 to $103.8 \text{ km s}^{-1} \text{Mpc}^{-1}$, and raises the high-tail probability from

$$P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.017 \quad (\text{baseline})$$

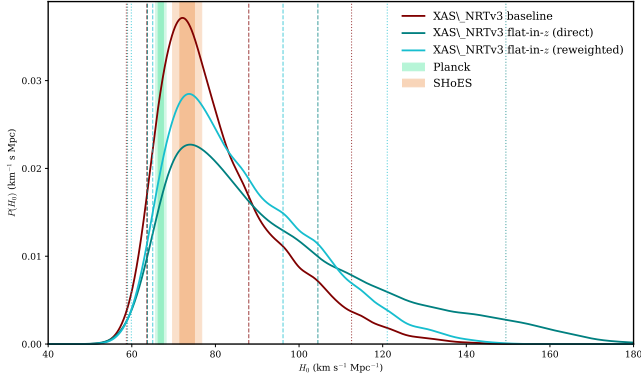
to

$$P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.159 \quad (\text{direct flat-in-z}),$$

and from 0.000 to 0.038 in $P(H_0 > 150 \text{ km s}^{-1} \text{Mpc}^{-1})$. The reweighted estimate captures only a small fraction of this shift:

Table 3. IMRPhenomXAS_NRTidalv3 H_0 summaries for the four prior variants on the default-mass full-sky prior set, $n_{\text{live}} = 5000$. MAP and sample-derived HPDs in $\text{km s}^{-1} \text{Mpc}^{-1}$. The reweighted variant is post-hoc and has no independent $\ln Z$.

Variant	MAP	68% HPD	$P(H_0 > 120)$	$P(H_0 > 150)$	$\ln Z$
Baseline ($\pi(d_L) \propto d_L^2$)	73.5	[63.8, 87.6]	0.017	0.000	486.25 ± 0.11
Flat-in- z , direct	71.9	[64.2, 103.8]	0.159	0.038	487.30 ± 0.10
Flat-in- z , reweighted	73.5	[65.2, 95.9]	0.041	0.000	(post-hoc)
$\sigma_{v_p} = 250 \text{ km s}^{-1}$	71.9	[61.7, 90.5]	0.069	0.015	485.55 ± 0.09

**Figure 4.** Prior-sensitivity comparison for IMRPhenomXAS_NRTidalv3. Weighted step-histograms of H_0 under the volumetric baseline and the two flat-in-redshift variants of Section 2.4, with the GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work’s standard-siren likelihood overlaid as the literature reference. The directly sampled flat-in-redshift posterior places substantially more mass at high H_0 than the reweighted estimate. The $\sigma_{v_p} = 250 \text{ km s}^{-1}$ peculiar-velocity sweep is reported in Table 3 and discussed inline rather than promoted to this figure.

$P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.041$, equivalent to

$$\frac{0.041 - 0.017}{0.159 - 0.017} \approx 17\% \quad (3)$$

of the direct prior-induced change. The 95 per cent upper bound under reweighting only reaches $95.9 \text{ km s}^{-1} \text{Mpc}^{-1}$, far short of the directly sampled $103.8 \text{ km s}^{-1} \text{Mpc}^{-1}$ at 68 per cent.

The four IMRPhenomXAS_NRTidalv3 evidence values agree to $\Delta \ln Z \lesssim 1.8$ – not decisive on the Jeffreys scale, and within the run-to-run variability seen in the public repository for this configuration. The data therefore do not prefer one distance prior over another at any meaningful significance: the change in H_0 across the prior variants is a *prior* effect, not a data-driven update. This is the central science finding of the paper.

4.2.0.1 Comparison with Abbott et al. (2017). The original bright-siren analysis used IMRPhenomPv2_NRTidal (precessing, NR-calibrated tides) and reported a minor change in the H_0 posterior under a flat-in-redshift reweighting of the volumetric-prior samples (Abbott et al. 2017b). Our locked primary, IMRPhenomXAS_NRTidalv3, replaces the precessing component with an aligned-spin base and uses the most recent NR-calibrated tidal phase prescription (NRTidalv3); IMRPhenomD_NRTidalv2, our anchor, uses the older NRTidalv2 calibration on the same aligned-spin base. The two model choices bracket the tidal-calibration axis of the original LVK waveform from the modern aligned-spin side. Repeating the four-variant prior sweep on IMRPhenomD_NRTidalv2 gives qualitatively the same behaviour as IMRPhenomXAS_NRTidalv3, with a larger amplitude: baseline MAP $71.6 \text{ km s}^{-1} \text{Mpc}^{-1}$, $P(H_0 >$

$120) = 0.076$; flat-in- z direct MAP $75.2 \text{ km s}^{-1} \text{Mpc}^{-1}$, $P = 0.281$; reweighted MAP $74.1 \text{ km s}^{-1} \text{Mpc}^{-1}$, $P = 0.195$; $\sigma_{v_p} = 250 \text{ km s}^{-1}$ MAP $72.3 \text{ km s}^{-1} \text{Mpc}^{-1}$, $P = 0.067$. The reweighting-capture fraction is $(0.195 - 0.076)/(0.281 - 0.076) \approx 58\%$, against $\approx 17\%$ for IMRPhenomXAS_NRTidalv3 (equation 3). The newer NRTidalv3 calibration in IMRPhenomXAS_NRTidalv3 tightens the upper H_0 tail relative to NRTidalv2 in IMRPhenomD_NRTidalv2 (Section 4.1: 95 per cent upper bounds 117.6 versus $130.0 \text{ km s}^{-1} \text{Mpc}^{-1}$), which leaves less Mode-B mass for the prior shift to redistribute and therefore both reduces the absolute prior-induced change in $P(H_0 > 120)$ and reduces the fraction of that change that reweighting can recover. The qualitative conclusion – that reweighting systematically understates the prior-induced change in the high- H_0 tail – is therefore robust across the tidal-calibration axis, and is in fact *more* severe under the modern waveform than under the older one. Under the original LVK waveform IMRPhenomPv2_NRTidal, where the precessing degrees of freedom further broaden the posterior, we expect the capture fraction to sit between the IMRPhenomXAS_NRTidalv3 and IMRPhenomD_NRTidalv2 values; a like-for-like reproduction would require porting that waveform to JAX and is left for future work.

4.2.0.2 Peculiar-velocity sweep. The $\sigma_{v_p} = 250 \text{ km s}^{-1}$ variant shifts the IMRPhenomXAS_NRTidalv3 H_0 posterior only slightly: H_0^{MAP} falls to $71.9 \text{ km s}^{-1} \text{Mpc}^{-1}$ and $P(H_0 > 120)$ moves from 0.017 to 0.069. For GW170817 with NGC 4993 identified, the dominant prior-induced uncertainty in H_0 is therefore the distance prior, not the peculiar-velocity uncertainty.

We caution against over-interpreting these shifts as physical results. None of them changes the qualitative statement that GW170817 alone is consistent with both early- and late-Universe H_0 measurements. The methodological point is that the magnitude of the shift under a change of distance prior is a property of the inference, not of the cosmology, and that the magnitude one would report using reweighting alone is materially different from the magnitude recovered by direct sampling.

4.3 The d_L - ι bimodality

The flat-in-redshift posterior is broad and asymmetric, with non-negligible weight at $d_L \lesssim 25 \text{ Mpc}$. To resolve whether this is a single broadened mode or a genuine bimodality, we run two prior-restricted variants of the direct flat-in-redshift analysis with IMRPhenomD_NRTidalv2 at $n_{\text{live}} = 5000$: a Mode-A run with $d_L \in [30, 75] \text{ Mpc}$ and a Mode-B run with $d_L \in [10, 30] \text{ Mpc}$. We retain the original GWTC-1 heterodyne reference, and additionally run an unrestricted $d_L \in [10, 75] \text{ Mpc}$ flat-in-redshift analysis with the heterodyne reference re-anchored at Mode B ($d_L = 20 \text{ Mpc}$, $\iota = 2.0 \text{ rad}$) to test whether the apparent Mode-B mass is a heterodyne-reference artefact.

Figure 5 shows the joint (d_L, ι) posterior under the unrestricted run, together with the per-mode H_0 marginals. The support is a curved

arm running from $(d_L, i) \approx (15, 1.9)$ Mpc-rad to $(40, 2.7)$ Mpc-rad with two distinct local maxima. Table 4 reports the local evidences and per-mode H_0 summaries.

The Bayes factor between Mode B and Mode A, after correcting for the volume difference between the restricted priors, is

$$\ln \mathcal{B}_{B/A} = (\ln Z_B - \ln Z_A) + \ln(20/45) = +0.15 - 0.81 = -0.66. \quad (4)$$

Mode B is therefore only mildly disfavoured by the data and contributes a non-negligible share of the joint posterior under any prior that gives the low- d_L region appreciable weight. The Mode-B-anchored unrestricted run recovers $P(H_0 > 120) = 0.281$ – statistically indistinguishable from the GWTC-1-anchored direct flat-in-redshift run reported in Section 4.2 – so the heterodyne-reference choice does not bias the Mode-A/Mode-B weight ratio.

This is the mechanism behind the reweighting capture fraction of equation 3. Mode B is the high- H_0 , small- d_L branch of the distance–inclination degeneracy. The baseline volumetric prior $\pi(d_L) \propto d_L^2$ assigns Mode B a tiny prior mass (the $[10, 30]$ Mpc slab carries less than 5% of the total $[10, 75]$ Mpc volumetric prior), and the corresponding region of the baseline posterior is sparsely sampled. Reweighting cannot reconstruct what is not there. Direct sampling under the flat-in-redshift prior populates Mode B at sampler resolution, and it is precisely the missing Mode-B mass that opens the gap between the directly sampled and reweighted estimates of $P(H_0 > 120)$. Posterior-coverage diagnostics that compare the baseline samples against the target prior should therefore be standard practice before reporting a reweighted bright-siren H_0 summary. The smaller capture fraction of IMRPhenomXAS_NRTidalv3 relative to IMRPhenomD_NRTidalv2 reflects exactly this: IMRPhenomXAS_NRTidalv3’s tighter upper tail leaves less Mode-B mass to be “rescued” by a prior shift in the first place.

4.4 Inline robustness summary

We complete the multi-axis robustness picture with sweeps that we ran but do not display as primary figures, since the central science finding of Section 4.2 is unaffected by any of them.

- *Sampler hyperparameters.* On the IMRPhenomD_NRTidalv2 default-mass baseline at $n_{\text{live}} = 5000$, varying $n_{\text{delete}}/n_{\text{live}} \in \{0.10, 0.25, 0.50, 0.75\}$ moves H_0^{MAP} by $\leq 1.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\ln Z$ by ≤ 0.6 ; varying the heterodyne-bin count $n_{\text{bins}} \in \{251, 501, 1001\}$ moves H_0^{MAP} by $\leq 0.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with the largest pairwise Wasserstein-1 distance between the three H_0 posteriors below $2 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

- *PSD source.* TaylorF2 runs at $n_{\text{live}} = 5000$ on the baseline prior with GWTC-1, kazewong, and BILBY reference PSDs span $H_0^{\text{MAP}} = 76.4\text{--}77.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $P(H_0 > 120) = 0.043\text{--}0.065$, again with maximum pairwise $W_1 < 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

- *Heterodyne reference.* Replacing the GWTC-1 reference parameters with the maximum-likelihood point of an initial low-cost optimisation produces IMRPhenomD_NRTidalv2 posteriors that are close to indistinguishable through the bulk of the distribution and depart only marginally in the high- H_0 tail.

All three axes therefore produce H_0 shifts at least an order of magnitude smaller than the prior-induced shift documented in Section 4.2.

5 PERFORMANCE AND RUNTIME STUDIES

5.1 Wall-clock runtime

The primary heterodyned GW170817 H_0 analysis with IMRPhenomXAS_NRTidalv3 and $n_{\text{live}} = 5000$ on the LVK-bounds prior set completes in ≈ 13 min on a single NVIDIA A100 GPU; the corresponding IMRPhenomD_NRTidalv2 run takes ≈ 14 min and the TaylorF2 run ≈ 9 min. The full four-variant prior-sensitivity suite (baseline, direct flat-in- z , reweighted post-hoc, $\sigma_{v_p} = 250 \text{ km s}^{-1}$) for IMRPhenomXAS_NRTidalv3 on the default-mass prior set fits inside an hour. The GW150914 IMRPhenomXPHM validation run takes ≈ 2 h on the same hardware – longer than an aligned-spin reference, and the price for the like-for-like LVK reproduction of Section 3.1.

5.2 Heterodyne speedup and live-point scaling

The heterodyned likelihood is the enabling approximation. Figure 6 compares heterodyned and unheterodyned IMRPhenomD_NRTidalv2 wall-clock time on GW170817 across $n_{\text{live}} \in \{500, 1000, 2500, 5000, 10\,000, 20\,000, 50\,000, 100\,000\}$ for the heterodyned sampler and $n_{\text{live}} \in \{500, 1500, 2500\}$ for the unheterodyned reference, plus the TaylorF2 unheterodyned point at $n_{\text{live}} = 1500$. At matched n_{live} the speedup factors are

$$\text{speedup}(n_{\text{live}}) = \begin{cases} 31.1 \times & n_{\text{live}} = 500, \\ 50.7 \times & n_{\text{live}} = 1500, \\ 67.7 \times & n_{\text{live}} = 2500. \end{cases} \quad (5)$$

The relative-binning approximation accounts for the bulk of the wall-clock reduction; the GPU saturation provided by BLACKJAX-NS is layered on top.

The heterodyned H_0^{MAP} is stable across $n_{\text{live}} \geq 2500$ to within $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $\ln Z$ stabilises within ± 0.2 over the same range. The $n_{\text{live}} = 5000$ configuration adopted for the science runs is therefore in the converged regime. Live-point counts up to 10^5 remain accessible on a single GPU; the $n_{\text{live}} = 10^5$ heterodyned run completes in roughly four hours.

5.3 Scope of the performance claim

We deliberately do not compare runtime against a single-core CPU baseline, which would give a misleadingly large speedup factor, nor against a production PARALLEL_BILBY run that we have not executed under matched settings. A like-for-like GW170817 PARALLEL_BILBY benchmark using IMRPhenomXAS_NRTidalv3 with identical priors and live-point count is the appropriate CPU comparison and is left for future work. The claim we make is narrower and reproducible: on a single A100 GPU, the full $n_{\text{live}} = 5000$ H_0 analysis completes in under a quarter of an hour, the three-waveform cross-check fits inside an hour, the live-point convergence study extends to $n_{\text{live}} = 10^5$ within an overnight budget, and the sky-prior matched-pair sweep takes well under a working day. This runtime budget is what makes the multi-axis robustness study scientifically routine rather than exceptional.

6 DISCUSSION

6.1 Implications for bright-siren cosmology

The main methodological message is that posterior reweighting, although statistically consistent in the limit of infinite baseline sam-

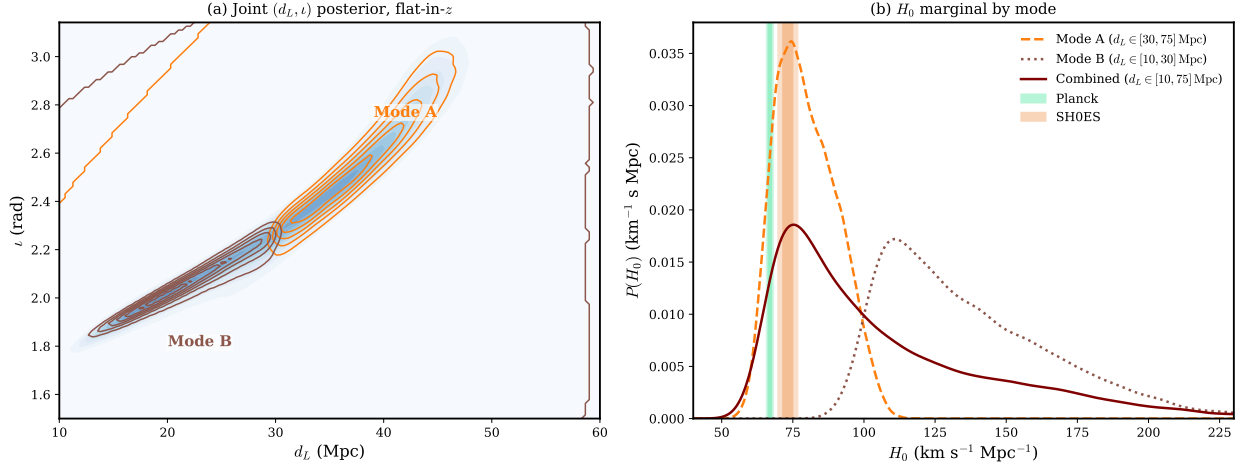


Figure 5. The d_L – ι bimodality in the GW170817 flat-in-redshift posterior. Left: joint (d_L, ι) posterior from the unrestricted $d_L \in [10, 75]$ Mpc run (Mode-B-anchored heterodyne reference), with the prior-restricted Mode-A and Mode-B contours overlaid. Right: per-mode H_0 marginals as weighted step-histograms. The Planck Λ CDM and SH0ES distance-ladder bands are shown for context; this work is itself a GW170817 reanalysis, so the LVK GW170817 band is uninformative here and is not included.

Table 4. Local evidences, MAP H_0 values, and sample-derived 68 per cent HPDs for the three d_L -restricted flat-in-redshift runs. The Mode-B/Mode-A Bayes factor includes a $\ln(20/45)$ correction for the difference in restricted-prior d_L volumes.

Variant	d_L range (Mpc)	H_0^{MAP} ($\text{km s}^{-1} \text{Mpc}^{-1}$)	68% HPD	$P(H_0 > 120)$	$\ln Z$
Mode A	[30, 75]	74.4	[66.4, 88.6]	$\sim 10^{-7}$	486.80 ± 0.10
Mode B	[10, 30]	110.8	[98.6, 152.1]	0.638	486.95 ± 0.09
Unrestricted	[10, 75]	75.2	[61.5, 118.0]	0.281	486.48 ± 0.09

ples, can substantially understate the prior sensitivity of bright-siren H_0 measurements in practice. For GW170817 under IMRPhenomXAS_NRTidalv3, switching from a volumetric distance prior to a flat-in-redshift prior changes $P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1})$ by a factor of ≈ 9 under direct sampling, while reweighting captures only $\approx 17\%$ of that shift (equation 3). The corresponding capture fraction on IMRPhenomD_NRTidalv2 – where the high- H_0 tail is heavier – rises to $\approx 58\%$. The deficit is a problem of effective sample coverage rather than of statistical methodology: the baseline volumetric prior assigns Mode B a small prior mass and reweighting cannot reconstruct what was not sampled.

For a single event this under-estimation does not change the cosmological interpretation: the GW170817 posterior is broad enough that both early- and late-Universe H_0 values lie within its 68 per cent HPD under any of the priors considered here. However, the scale of the effect matters for forecasts and combined analyses. When bright-siren posteriors are combined over multiple events, or when tail probabilities are compared against external priors from cosmological surveys, the prior-dependence of each single-event posterior propagates through. An analysis that quantifies that dependence only through reweighting risks a systematic bias in the direction of under-estimating the prior contribution.

6.2 Where reweighting is sufficient

Reweightings are adequate when the target and baseline priors have similar support in the regions that carry most of the posterior mass. The $\sigma_{v_p} = 250 \text{ km s}^{-1}$ variant in our analysis is a case of this kind: the change modifies the host-velocity likelihood smoothly over the scale of the velocity posterior, the baseline samples cover the affected region well, and the reweighted and directly sampled σ_{v_p} posteriors

agree to well within the statistical width. The flat-in-redshift case fails the same coverage test, because the baseline volumetric prior and the target flat-in-redshift prior differ most at the edges of the d_L range. The mode-isolated runs of Section 4.3 make this concrete: Mode B has $\ln \mathcal{B}_{B/A} = -0.66$ relative to Mode A under the flat-in-redshift prior, but its baseline-volumetric-prior mass is below 5% of Mode A’s, so reweighted samples are essentially blind to it. Diagnostics such as an effective-sample-size ratio under the reweighted weights, or a direct overlay of the d_L marginals, can flag such cases before the reweighted summary is reported.

6.3 Implications for future bright sirens

The practical consequence is that direct prior sampling should be used for bright-siren H_0 analyses whenever the target prior differs non-trivially in support from the baseline, rather than being used only as a cross-check. The runtime budget on a single A100 GPU now permits such reruns routinely (Section 5); the live-point and sampler-robustness studies show that the science-quality configuration sits in the converged regime, so the marginal cost of a prior-sensitivity rerun is the marginal cost of one ~ 15 min run. Live-point counts up to 10^5 are accessible on the same hardware, which means that prior-sensitivity studies can in principle be combined with population-level analyses without a substantial increase in end-to-end wall-clock time. This is directly relevant to the third-generation detector era (Hu & Veitch 2025), where the detection rate of compact-binary mergers will make repeated, controlled posterior reanalyses the norm rather than the exception.

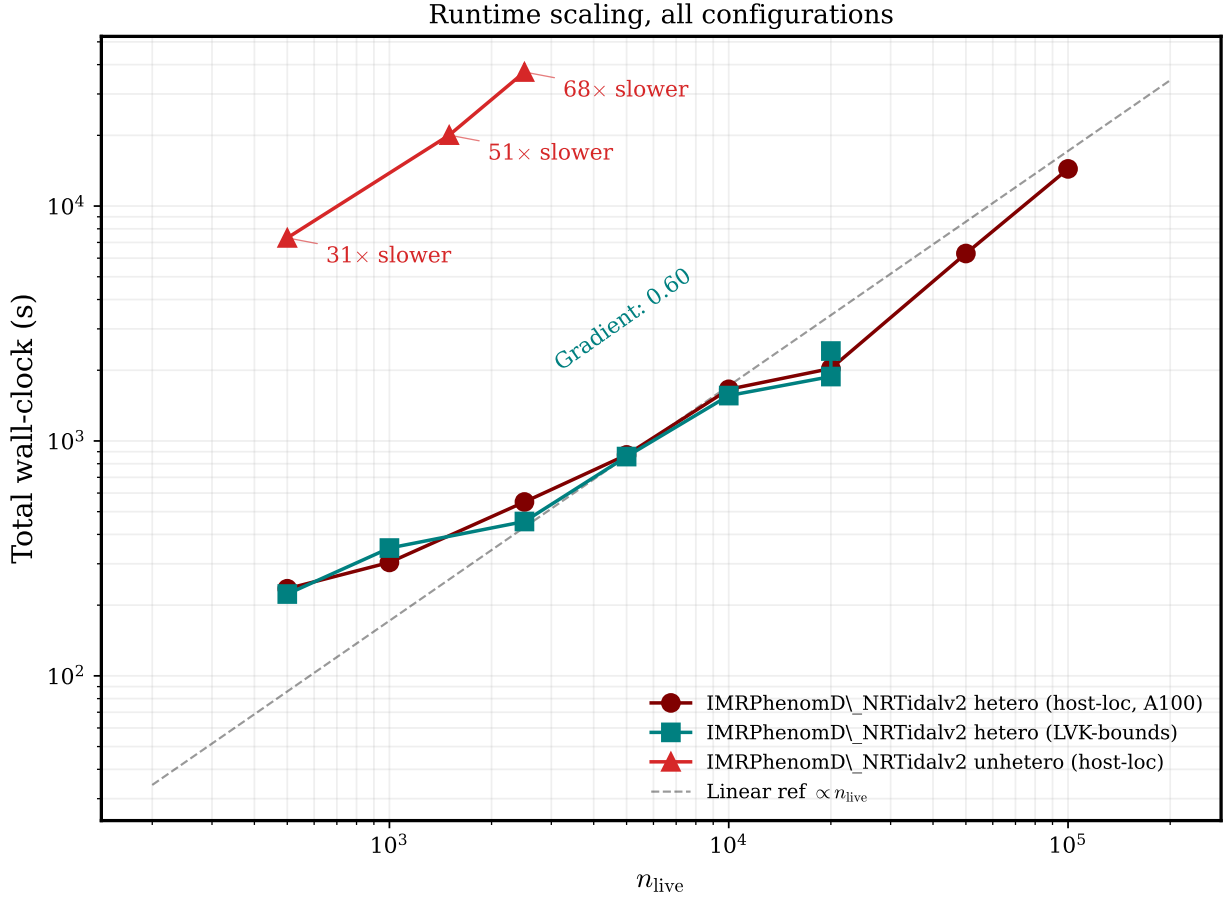


Figure 6. GW170817 wall-clock scaling. Left: heterodyned and unheterodyned wall-clock as a function of n_{live} for the IMRPhenomD_NRTidalv2 family on default-mass and LVK-bounds prior sets, with the TaylorF2 unheterodyned point at $n_{\text{live}} = 1500$ for context. Right: heterodyne-vs-unheterodyne speedup factor at matched n_{live} , with the 31×/51×/68× values annotated.

6.4 Limitations and open items

Several limitations should be acknowledged. First, no waveform in our JAX inventory simultaneously includes precession and tides; the locked primary is therefore the aligned-spin tidal IMRPhenomXAS_NRTidalv3, and the precession contribution is bracketed inline by the IMRPhenomPv2 (no tides) cross-check rather than by a preprocessing tidal model. A like-for-like reproduction of the LVK IMRPhenomPv2_NRTidal analysis would require porting that waveform to JAX. Second, the host-galaxy association with NGC 4993 and the peculiar-velocity inputs are taken directly from Abbott et al. (2017b); we have varied σ_{v_p} but not $\langle v_p \rangle$. Third, we do not present a matched PARALLEL_BILBY benchmark; the performance claim is therefore explicitly restricted to single-A100 wall-clock runtime on the current pipeline. Fourth, the d_L - ι bimodality is characterised at the level of local evidences and mode-isolated marginals; we do not at present propagate Mode B into a population analysis or combine it with an electromagnetic likelihood for the host inclination.

7 CONCLUSIONS

We have used a GPU-native nested-sampling pipeline to revisit the GW170817 bright-siren H_0 measurement of Abbott et al. (2017b) with the locked primary aligned-spin tidal waveform IMRPhenomXAS_NRTidalv3, focused on whether the post-hoc prior-

reweighting step in that analysis recovers the answer that direct sampling under the target prior would give. On the LVK-bounds prior set, IMRPhenomXAS_NRTidalv3 yields $H_0^{\text{MAP}} = 72.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ with sample-derived 68 per cent HPD $[63.1, 88.5] \text{ km s}^{-1} \text{ Mpc}^{-1}$; IMRPhenomD_NRTidalv2 and TaylorF2 give consistent values within their HPDs, and the GWTC-1 IMRPhenomPv2_NRTidal posterior mapped through this work’s standard-siren likelihood peaks at $67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ within the same envelope. The four-variant prior-sensitivity sweep on the default-mass full-sky prior set has baseline $H_0^{\text{MAP}} = 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $P(H_0 > 120 \text{ km s}^{-1} \text{ Mpc}^{-1}) = 0.017$; switching the distance prior to flat-in-redshift by direct sampling raises that tail probability to 0.159, while post-hoc reweighting captures only 0.041 – approximately 17% of the shift. The four IMRPhenomXAS_NRTidalv3 evidence values agree to $\Delta \ln Z \lesssim 1.8$, so the data do not prefer one distance prior over another: the H_0 shift is a prior effect, not a data-driven update. The mechanism is the d_L - ι bimodality: Mode A ($d_L \in [30, 75] \text{ Mpc}$, $H_0^{\text{MAP}} = 74.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and Mode B ($d_L \in [10, 30] \text{ Mpc}$, $H_0^{\text{MAP}} = 110.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$) carry comparable local evidence after prior-volume normalisation ($\ln \mathcal{B}_{B/A} = -0.66$); reweighting cannot recover Mode B because the baseline volumetric prior assigns it negligible mass.

The heterodyned likelihood gives wall-clock speedups of 31×, 51×, and 68× over the unheterodyned reference at $n_{\text{live}} = 500, 1500, 2500$ on a single A100 GPU. A matched-pair sky-prior

sweep at $n_{\text{live}} = 5000$ confirms that imposing a ± 0.05 rad patch around NGC 4993 changes wall-clock by only $0.86\times$ for the heterodyned IMRPhenomD_NRTidalv2 sampler, $0.98\times$ for unheterodyned IMRPhenomD_NRTidalv2, and $1.44\times$ for unheterodyned TaylorF2, so the speedup is not driven by a tight sky prior. The scientific value of fast nested sampling for bright-siren cosmology is that direct prior reruns, multi-waveform cross-checks, and full sampler-robustness studies all become routine, replacing post-hoc reweighting and isolated cross-checks as the default robustness tool without a substantial change in the computational budget.

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DATA AVAILABILITY

This paper uses publicly available GW170817 and GW150914 strain data and GWTC-1/GWTC-2.1 reference posteriors from the Gravitational Wave Open Science Center (GWOSC). All derived posterior-sample CSV files, summary tables, scaling logs, and figure-generation scripts are distributed with the project repository at [TODO:insertpermanentrepositoryURL/DOI](#), with provenance recorded in `paper_knowledge_base/result_link_index.md` and the scaling-study summary CSV. TODO: confirm GWOSC release identifiers for the GW170817 and GW150914 strain segments before submission.

REFERENCES

- Abbott B. P., et al., 2016, *Physical Review Letters*, 116, 061102
 Abbott B. P., et al., 2017a, *Physical Review Letters*, 119, 161101
 Abbott B. P., et al., 2017b, *Nature*, 551, 85
 Ashton G., et al., 2019, *The Astrophysical Journal Supplement Series*, 241, 27
 Bradbury J., et al., 2018, JAX: composable transformations of Python+NumPy programs, <https://github.com/google/jax>
 Cabezas A., et al., 2024, BlackJAX: Composable Bayesian inference in JAX
 Cornish N. J., 2013, Heterodyned likelihood for rapid gravitational-wave parameter estimation ([arXiv:1307.4820](#))
 Edwards T. D. P., et al., 2023, Ripple: Differentiable and hardware-accelerated gravitational waveforms
 Hu Q., Veitch J., 2025, Computational costs of gravitational-wave parameter estimation in the third-generation detector era
 Krishna K., et al., 2023, Accelerated parameter estimation with relative binning in Bilby ([arXiv:2312.06009](#))
 Planck Collaboration 2016, *Astronomy and Astrophysics*, 594, A13
 Prathaban M., et al., 2025, Gravitational-wave inference at GPU speed: A bilby-like nested sampling kernel within blackjax-ns ([arXiv:2509.04336](#))
 Riess A. G., et al., 2016, *The Astrophysical Journal*, 826, 56
 Riess A. G., et al., 2022, *The Astrophysical Journal Letters*, 934, L7
 Schutz B. F., 1986, *Nature*, 323, 310
 Skilling J., 2006, *Bayesian Analysis*, 1, 833
 Speagle J. S., 2020, *Monthly Notices of the Royal Astronomical Society*, 493, 3132
 Wong K., et al., 2023, Jim: A JAX-based gravitational-wave inference library

- Wouters T., et al., 2024, Accelerated gravitational-wave inference for binary neutron stars with Jim ([arXiv:2404.11397](#))
 Yallup D., et al., 2025, GPU nested slice sampling for gravitational-wave inference ([arXiv:2509.24949](#))
 Zackay B., Dai L., Venumadhav T., 2018, Relative binning and fast likelihood evaluation for gravitational-wave parameter estimation ([arXiv:1806.08792](#))